

When Does Network Diversity Reflect Physical Diversity?

A Cross-Layer Audit of Internet Service Paths

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Abstract—Network-layer path diversity is often used as a proxy for the physical diversity supporting Internet services. We present CLASP, a Cross-Layer Audit of Service Paths that compares network-transition and feasible landing-region-corridor distributions over the same traceroute segments. We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering all 13 DNS Roots, three applications, and two dynamic multi-target topology references, yielding 370 auditable country-measurement units. Under equal-share allocation, 81 of 275 service-facing units have broad network support and concentrated corridor support, while 158 remain broad at both layers; effective category count contracts in 204 units (74.2%). Across units, the Spearman association between network and corridor concentration, termed Layer Agreement, varies by geography. Under projection-score weighting it is 0.632 for 67 island and archipelagic units and 0.001 for 202 coastal-mainland units. The island-coastal difference remains positive under equal-share, projection-score, and Top-1 allocation, at 0.461, 0.631, and 0.715. These results show why auditing physical-corridor diversity requires both measuring within-unit cross-layer differences and examining how concentration rankings correspond across geographic populations.

Index Terms—Internet measurement, submarine cables, traceroute, cross-layer analysis, anycast, path diversity

I. INTRODUCTION

Submarine cables form the physical backbone of the global Internet and carry nearly all intercontinental traffic [1]. Cable failures can isolate regions, degrade remote-service access, and redirect traffic toward longer or congested alternatives. Routes that share a cable or coastal corridor can be affected by the same physical event even when their network-layer paths appear different. Evaluating path diversity therefore requires measuring whether logically distinct routes also occupy distinct physical directions.

Path diversity is usually measured at the network layer. Traceroute records routers and Autonomous System (AS) transitions, so distinct hops or AS paths are counted as distinct alternatives. It does not identify the submarine infrastructure that could carry those paths. Two paths that appear independent in traceroute may leave a region through the same coast and rely on the same small set of feasible corridors. Whether network-layer diversity corresponds to physical-corridor diversity is therefore an empirical question.

Prior work supplies the cross-layer mappings needed to study this question. Internet Atlas, InterTubes, and iGDB connect observed Internet links to long-haul infrastructure; Nautilus associates IP links with candidate submarine

cables; Calypso maps routes using cable layouts, network relationships, and inland connectivity; and Xaminer applies cross-layer maps to infrastructure risk analysis [2]–[7]. These systems identify physical infrastructure that could carry an individual link or path. We use such assignments to measure a population-level property: whether concentration among observed network transitions tracks concentration among feasible physical corridors.

We present CLASP, a Cross-Layer Audit of Service Paths. Within each country-measurement population, CLASP constructs a network-transition distribution and a feasible landing-region-corridor distribution from the same traceroute segments. This paired construction supports two comparisons: changes in concentration and breadth within a unit, and the association between concentration rankings across units. We call the latter *Layer Agreement*, measured by Spearman rank correlation.

Three questions organize the measurement. **RQ1** asks how network and feasible-corridor concentration and breadth differ within the same country-measurement population. **RQ2** asks whether network concentration tracks corridor concentration across units, and how this association varies by geography and candidate allocation. **RQ3** asks how exposure, absolute corridor breadth, and measurement-family composition contextualize these cross-layer results.

We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering all 13 DNS Roots, Wikipedia, Reddit, Netflix Assets, and two dynamic multi-target topology references. The aligned one-hour snapshot yields 370 auditable country-measurement units: 222 DNS Root, 53 application, and 95 topology-reference units. Each unit retains its source-country and measurement identity.

Across service-facing units, broad network support can co-exist with concentrated corridor support. Under equal-share allocation, 81 of 275 units (29.5%) are network-broad/corridor-concentrated, while 158 (57.5%) remain broad at both layers. Eighteen units are concentrated at both layers, and 18 expand from concentrated network support to broad corridor support. Continuous measures reveal further differences: effective category count contracts in 204 units (74.2%). The paired audit thus identifies both changes in concentration class and narrowing within the same class.

The relationship between the two concentration rankings varies by geography. Under projection-score weighting, Layer Agreement is 0.632 among 67 island and archipelagic service-

facing units from 10 countries and 0.001 among 202 coastal-mainland units from 28 countries. Network concentration tracks corridor concentration more closely in the island and archipelagic group; the coastal-mainland group exhibits almost no monotonic rank association. The between-group difference is 0.631.

The geographic ordering persists under all three candidate-allocation rules. Equal-share allocation gives Layer Agreement of 0.380 for island and archipelagic units and -0.081 for coastal-mainland units; projection-score weighting gives 0.632 and 0.001; and Top-1 gives 0.820 and 0.105. The corresponding differences are 0.461, 0.631, and 0.715. These results connect the two levels of the audit: within-unit changes reveal where network and corridor diversity differ, while Layer Agreement describes how closely their rankings track across geographic populations.

This paper makes three contributions. First, CLASP constructs paired network-transition and feasible-corridor distributions over identical traceroute segments, with trace-level exposure reported separately. Second, the audit quantifies within-unit cross-layer differences and identifies a geographic contrast in concentration-rank association that persists under three candidate-allocation rules. Third, the reproducible audit records measurement IDs, configuration values, mapping states, and pipeline counts so that the reported distributions can be reconstructed and inspected.

II. BACKGROUND AND RELATED WORK

A. Reading Service Paths at the Network Layer

Users reach services through domain names, CDN resources, replicas, and anycast targets. Routing selects the resulting path, and traceroute exposes the intervening IP hops and AS transitions. These transitions are the usual unit of path-diversity measurement: they are counted, compared across vantage points, and used to describe how many distinct routes a service presents.

This is also the layer at which service reachability is commonly assessed. Prior work measures the user’s view of submarine criticality [8], characterizes anycast deployment and its network footprint [9], [10], and relates cable infrastructure to AS-level structure [11]. These measurements describe network-layer reachability. Our question is whether diversity at that layer corresponds to breadth across feasible physical directions.

The relationship is many-to-many. Different routers or AS transitions may project onto one landing-region corridor, while a recurring network transition may support several physical alternatives. A *landing-region corridor* is a direction-independent pair of bounded coastal regions connected by at least one feasible cable candidate. Exact landing stations preserve facility detail but split nearby access points serving the same direction; country-level inventories merge coastlines and international directions. Landing regions retain coastal structure between these two resolutions. Corridors are the appropriate unit for comparing physical directions: exact cable

identity is often ambiguous, and counting several candidate cables through nearby landings as independent directions would overstate the alternatives. The aggregation preserves distinct coastal exit directions while consolidating parallel candidate systems that represent the same direction.

B. Mapping Paths onto Submarine Infrastructure

Event-oriented studies connect new submarine infrastructure to observed routing changes. Bischof et al. characterized Cuba’s connectivity before and after ALBA-1 and later formulated the task of connecting Internet observations to the worldwide cable mesh [12], [13]. Fanou et al. measured routing changes following new cable deployments [14]. Liu et al. combine resource discovery, RIPE Atlas traceroutes, geolocation, and propagation feasibility to quantify how users reach Web resources through submarine paths [8].

Cross-layer cartography supplies the assignments used by our measurement. Internet Atlas and InterTubes associate long-haul facilities with observed paths [2], [3]. iGDB integrates facilities, fiber, and network entities into a cross-layer map [4]. Nautilus associates IP links with candidate submarine cables and attaches confidence to each assignment [5]. Calypso combines cable layouts, network relationships, latency, and inland connectivity to map routes and defines Route Stress over those assignments [6]. Xaminer applies cross-layer maps to disaster-oriented infrastructure analysis [7]. Measurements of intercontinental links also identify layer-3 links and gateway routers [15].

These systems produce an assignment from a link or path to physical infrastructure that could plausibly carry it. Their validation evaluates the assignment using cable failures, targeted traceroutes, operator maps, or expert feedback. CLASP adopts this projection approach rather than claiming to identify ground-truth cable use. The individual assignments are inputs to our measurement; distributions aggregated over a path population are the objects we compare.

C. From Mapping to Distribution Measurement

An assignment does not by itself describe how observation mass is distributed across physical directions or whether that distribution tracks the one visible at the network layer. We separate two quantities that are easy to conflate. *Submarine exposure* is the fraction of valid traceroutes containing a feasible inter-region corridor. *Corridor concentration* describes how candidate-bearing segment mass is distributed across corridors after entry. A country can enter the candidate space frequently and distribute observations widely, or enter infrequently with the exposed observations concentrated on one direction.

Country and measurement identity jointly define each observation population. Source geography and interconnection describe the context from which paths start; anycast, replica selection, and routing determine the destinations and paths observed during the measurement window. DNS Roots illustrate this interaction because functionally equivalent services operate independent anycast deployments. The three applications

retain their own selected destinations. The two dynamic multi-target measurements deliberately sample a broader target population and provide topology-reference distributions against which the service-facing distributions can be read.

CLASP aligns the statistical unit across layers. A network transition and its corridor support originate from the same atomic segment, so their concentration difference describes how a fixed observation population is organized. The framework next defines this paired construction, and Section IV records the aligned path and infrastructure inputs.

III. CLASP: THE PAIRED CROSS-LAYER AUDIT

A. Overview

CLASP constructs paired network-transition and feasible-corridor distributions for each country-measurement unit. It combines traceroute, probe metadata, IP geolocation and ASN mappings, AS context, and cable metadata. Figure 1 shows the processing order: path extraction creates atomic hop-pair segments, corridor projection constructs each segment's feasible landing-region corridor set, and aggregation forms the two distributions over the same candidate-bearing segments.

The construction keeps trace-level exposure separate from segment-level concentration. Exposure records whether a valid traceroute contains at least one feasible inter-region corridor. The paired distributions condition on candidate-bearing segments and describe how their observation mass is organized. Because these quantities use different denominators, CLASP reports them separately.

B. Path Extraction

We group traceroutes by probe country and measured service. This grouping retains the source-side connectivity context and the destination selection observed for each measurement. Hop countries remain segment attributes rather than grouping keys.

Each traceroute is normalized into its complete sequence of visible hops with locations, ASNs, and RTTs. Successive visible, geolocated hops form atomic segments. Each segment records endpoint locations, ASNs, countries, hop positions, RTT difference, and bridged-hop count. Repeated hops and RTT samples retain their trace identity and ordering. When timeouts separate two visible hops, the bridged-hop count records the hidden gap for audit accounting.

When both endpoint ASNs are available, their ordered pair forms the segment's network-transition label. A segment missing one endpoint ASN receives an ordered country-pair fallback label, which is strictly coarser. We record the fallback share for each unit. Audit eligibility requires that share to be at most 30%, together with at least 30 candidate-bearing segments, 10 probes, and 3 probe ASNs. These thresholds remove units whose network labels or observation support are too sparse for the paired summaries; they are quality floors rather than guarantees of representativeness.

C. Feasible Landing-Region Corridors

a) *Landing stations and cable lifecycle.*: For each segment endpoint, CLASP retrieves landing stations within 50 km and pairs them through cable systems in service on July 1, 2026. The catchment allows endpoint locations to be associated with nearby coastal facilities at the resolution of the geolocation input. Explicit cable path or branch information defines direct landing-point adjacency. When the inventory does not provide that information, the pipeline retains reachability at the metadata's topology resolution and records that provenance state rather than presenting it as an observed cable traversal.

b) *Landing regions.*: Landing stations are grouped into regions with maximum pairwise distance 30 km. The diameter merges nearby facilities serving the same coastal direction while retaining separated coastal entry and exit patterns. A *landing-region corridor* is a direction-independent pair of regions connected by at least one feasible cable candidate. Several cable systems and exact landing pairs may support one corridor. Exact landing-pair, cable, and corridor identities remain linked in the outputs so that the aggregation can be audited at each resolution.

c) *Propagation feasibility.*: A candidate must satisfy geographic, lifecycle, and propagation constraints. Let D be the great-circle distance between its landing points, $v_f = 200$ km/ms the configured effective propagation speed in fiber, ΔRTT the segment's RTT difference, and $\tau = 5$ ms the RTT tolerance. The candidate is propagation-feasible when

$$\Delta RTT + \tau \geq \frac{2D}{v_f}. \quad (1)$$

The inequality asks whether the observed round-trip increment can accommodate the minimum round-trip propagation time between the landing points. When ΔRTT is non-positive or inconclusive, CLASP retains the geographic and lifecycle constraints and flags the segment. The 200 km/ms speed and 5 ms tolerance are fixed configuration values in the reported analysis.

D. Candidate Allocation and Observation Mass

Duplicate cable rows belonging to the same corridor are collapsed within an atomic segment. For a segment s with feasible corridor set $\mathcal{C}_s$, the primary analysis allocates its observation mass using the projection score:

$$w_{s,c} = \frac{\sigma_{s,c}}{\sum_{c' \in \mathcal{C}_s} \sigma_{s,c'}}, \quad \sum_{c \in \mathcal{C}_s} w_{s,c} = 1, \quad (2)$$

where $\sigma_{s,c}$ sums the evidence-support scores of the cable candidates assigned to corridor c . The implementation forms each candidate score as the product of landing-proximity support, AS-economic support, RTT support, and a latency penalty. Landing-proximity support decreases with the distances from the two hop endpoints to their candidate landing points. AS-economic support uses endpoint-AS membership in cable-owner AS sets and relationships or owner-group reachability in the AS graph. RTT support compares the observed

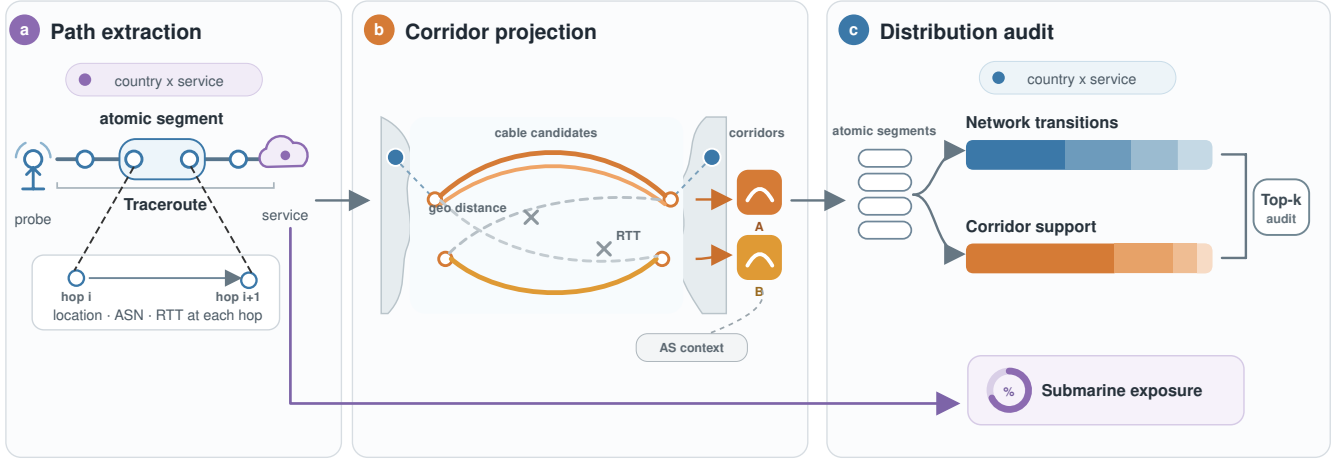


Fig. 1. CLASP extracts atomic segments, projects feasible landing-region corridors, and constructs paired network-transition and corridor distributions over the same candidate-bearing observations.

TABLE I
SEGMENT-LEVEL PHYSICAL MAPPING RESOLUTION.

State	Operational definition
Single corridor	Feasibility inputs available and $ C_s = 1$
Bounded multi-corridor	Feasibility inputs available and $ C_s > 1$
No feasible corridor	Feasibility inputs available and $ C_s = 0$
Insufficiently resolved	Insufficient location, path-visibility, or meta-data fields for stable corridor construction

increment with the propagation lower bound and applies a penalty to highly inflated short-path increments; inconclusive increments receive neutral RTT factors. The candidate scores are summed by corridor and normalized as in Eq. (2), so every candidate-bearing segment contributes one unit of mass.

Projection-score weighting retains the relative evidence used to rank feasible candidates and is the primary allocation for Layer Agreement. These weights express relative support within a candidate set. Endpoint-AS information contributes both to network-transition labels and to candidate ranking, which makes the equal-share comparison particularly useful: equal sharing assigns $w_{s,c} = 1/|C_s|$ to each feasible corridor without using the score magnitudes. Top-1 allocation places all mass on the highest-scoring corridor. Together, the three rules compare equal support, graded support, and a single preferred candidate over the feasible sets.

E. Mapping Resolution and Pipeline Accounting

Table I classifies each segment by input coverage and the size of its feasible corridor set. Single- and bounded multi-corridor segments form the physical distribution. Segments with no feasible corridor or insufficient inputs remain in pipeline accounting, and every valid traceroute remains in the exposure denominator.

Candidate-set size and aggregate concentration describe different levels. A unit may contain many bounded multi-corridor

segments yet concentrate after their allocated mass repeatedly supports the same corridors. CLASP therefore retains the resolution state instead of using candidate multiplicity as a substitute for distribution breadth.

F. Paired Distributions

For an auditable country–measurement unit u , let S_u denote its candidate-bearing atomic segments and $\ell(s)$ the ordered network label of segment s . The network-transition probability of label t is

$$p_u^N(t) = \frac{\sum_{s \in S_u} \mathbf{1}[\ell(s) = t]}{|S_u|}. \quad (3)$$

The corridor distribution sums the masses in Eq. (2) for each corridor and normalizes within the same S_u . Before normalization, both distributions have total mass $|S_u|$. Their difference therefore describes how the same candidate-bearing observations are organized at the two layers, not a comparison between different trace populations. Direction is preserved for network transitions; the physical corridor is direction-independent because it represents a coastal connection rather than forwarding order.

Submarine exposure uses the complete valid-trace population. A traceroute is exposed when at least one of its segments has a feasible inter-region corridor. The number of exposed traces divided by all valid traces in the unit gives the exposure rate. This denominator differs intentionally from $|S_u|$, which conditions the distributions on candidate-bearing segments.

G. Distribution Summaries and Layer Agreement

Three metrics summarize each distribution. Top-2 share is the sum of the two largest probabilities and records mass in the dominant categories. Effective category count is $\exp[-\sum_i p_i \log p_i]$ and reports breadth on the scale of an equally weighted category count. Normalized entropy divides Shannon entropy by the logarithm of observed support size

TABLE II
CLASP MEASUREMENT CONFIGURATION.

Parameter	Value
Endpoint landing catchment	50 km
Landing-region maximum diameter	30 km
Fiber propagation speed	200 km/ms
RTT tolerance	5 ms
Same-city threshold	25 km
Cable lifecycle date	July 1, 2026
Inconclusive RTT	Retained and flagged
Timeout-bridged segment	Retained and flagged
Path scope	Complete visible sequence

and measures evenness relative to that support. Network-to-corridor differences are computed within each unit.

Layer Agreement is the Spearman rank association, across a reported set of units, between network Top-2 share and corridor Top-2 share. A coefficient approaching +1 indicates closely aligned concentration rankings; a coefficient near zero indicates little monotonic rank association, and a negative coefficient indicates opposing rankings. This comparison concerns ordering across units. Within-unit Top-2 differences and effective category counts separately measure how concentration and breadth change across layers. A group can therefore exhibit aligned rankings alongside substantial within-unit narrowing. We report Layer Agreement by measurement family, geography, and candidate-allocation rule.

For the frozen equal-share result, we quantify geographic uncertainty at the level where geography is assigned. A cluster-bootstrap replicate samples countries with replacement within each geographic group and carries every country–measurement unit from each sampled country together. We use 50,000 replicates and report percentile intervals. A separate two-sided reference test permutes the geographic labels of whole countries 100,000 times while retaining the observed number of countries in each group. Both procedures use seed 20260919. This country-clustered analysis is applied to the frozen equal-share unit table; projection-score and Top-1 results provide the allocation-robustness comparison.

A common 80% Top-2 threshold supplies four descriptive classes: broad at both layers, concentrated at both, network-broad/corridor-concentrated, and network-concentrated/corridor-broad. Continuous Top-2 shifts, effective counts, and normalized entropy accompany the classes so that threshold crossings do not replace the underlying distributions.

IV. DATASETS

All inputs are aligned to July 1, 2026 (UTC). Network-path observations, IP and AS annotations, and cable lifecycle records therefore describe a common analysis period. We combine these observations with network and submarine-cable metadata.

A. RIPE Atlas Measurements

Our network observations comprise 18 public RIPE Atlas IPv4 traceroute measurements collected from 00:00 to

01:00 UTC. Thirteen measurements cover independently operated anycast DNS Root services. Wikipedia, Reddit, and `assets.nflxext.com` provide three application-facing observations. Measurements 5051 and 5151 use dynamic targets as multi-target UDP and ICMP topology references. Table III summarizes the corpus after canonical trace-identity deduplication.

Each record retains the destination address returned by RIPE Atlas. This preserves the multi-target structure of the two dynamic-target measurements and the destination instances selected by service measurements during the aligned window.

The measurement families do not form a controlled comparison. The two dynamic-target measurements sample many destinations during the window, whereas each service measurement retains the destinations selected for that service. We use the former as topology references and describe differences between the observed populations; we do not attribute those differences to service deployment or target selection.

B. Supporting Datasets

Table IV lists the supporting datasets and versions. RIPE Atlas probe metadata supplies the probe country and source-network context [16]. IPinfo Location and ASN databases annotate visible hops with geographic and AS information [17].

The Submarine Cable Map provides the cable and landing-point inventory, including lifecycle and organizational fields [1]. The analysis includes systems in service on the measurement date. CAIDA AS Relationships and the derived owner-AS associations supply supplementary ranking context [18]. A versioned country taxonomy classifies probe countries as island or archipelagic, landlocked, or other coastal.

V. RESULTS

The paired audit reveals cross-layer differences within country–measurement units and geographic variation in the association between their concentration rankings. We first quantify within-unit changes, then compare Layer Agreement across geographic groups and candidate-allocation rules. Exposure, absolute corridor breadth, and measurement-family summaries provide context for these results.

The processing pipeline retains 490,911 valid traceroutes from 707,314 raw records and yields 3,291,063 atomic segments. Of 2,432,559 mappable segments, 555,119 reach candidate landing points, 171,232 satisfy the corridor constraints, and 124,350 support at least one feasible inter-region corridor. The equal-share audit records 71,990 single-corridor and 52,360 bounded multi-corridor segments. After the eligibility filters in Section III-B, the analysis contains 370 auditable country–measurement units: 222 DNS Root, 53 application, and 95 topology-reference units.

A. Network and Corridor Diversity Differ within Units

Under equal-share allocation, the 80% Top-2 threshold identifies four service-facing cross-layer outcomes (Fig. 5). Among 275 service-facing units, 81 (29.5%) are network-broad/corridor-concentrated, 158 (57.5%) remain broad at both

TABLE III
RIPE ATLAS MEASUREMENT CORPUS.

Group	Target	MSM ID(s)	Raw	Valid
DNS Roots	A-M Root	5009, 5010, 5011, 5012, 5013, 5004, 5014, 5015, 5005, 5016, 5001, 5008, 5006	534,276	317,873
Applications	Wikipedia; Reddit; Netflix Assets	86710103; 176906957; 176517335	115,596	115,596
Topology references	IPv4 UDP; IPv4 ICMP	5051; 5151	57,442	57,442
Total	18 measurements	-	707,314	490,911

TABLE IV
SUPPORTING DATASETS.

Dataset	Version	Content
RIPE Atlas probe metadata	2026-07-01	Probe country and source ASN
IPinfo Location MMDb	2026-07-01	Hop geolocation
IPinfo ASN MMDb	2026-07-01	IP-AS mapping
Submarine Cable Map	2026-07-01	694 cables; 1,916 landing points
CAIDA AS Relationships	2026-07-01 release	AS relationships
Owner-AS mapping	2026-07-01 inputs	Owner-AS associations
Country geography taxonomy	2026-07-01	Operational geography classes

layers, 18 (6.5%) are concentrated at both, and 18 (6.5%) are network-concentrated/corridor-broad. Four of 95 topology-reference units (4.2%) enter the first class; the other 91 remain broad at both layers. The 81 units in the first class directly identify populations whose broad network support accompanies concentrated feasible-corridor support.

Continuous measures reveal narrowing within as well as across concentration classes. Across all service-facing units, corridor Top-2 share increases in 184 of 275 (66.9%), and effective category count contracts in 204 (74.2%). Of 239 units with a broad network distribution, 178 (74.5%) move upward in Top-2 share: 81 cross the threshold, while 97 remain broad at both layers with a median increase of 6.3 percentage points. Thus, remaining in the broad–broad class can coexist with an increase in concentration. DNS Roots and applications enter the network-broad/corridor-concentrated class at similar rates: 66 of 222 DNS units (29.7%) and 15 of 53 application units (28.3%). Their median continuous Top-2 shifts are 11.0 and 21.0 percentage points.

Three existing cases distinguish continuous narrowing, expansion, and changes in absolute breadth from support-relative evenness. **Singapore–Netflix** moves from 32.5% network Top-2 share to 65.0% corridor Top-2 share, while its effective count contracts from 19.86 to 3.41. The continuous measures capture this substantial narrowing within the broad–broad class. **New Zealand–Netflix** moves from 63.6% to 37.6%, with effective count expanding from 5.18 to 6.92. **Singapore–H-Root** moves from 39.6% to 50.1%, and its effective count contracts from 20.67 to 5.99 while normalized entropy changes only slightly.

Bounded multi-corridor segments account for 99.9%, 78.1%, and 98.2% of candidate-bearing observations in these three cases. Repeated overlap among the bounded sets produces different aggregate distributions under the same projection and aggregation rules. Together, the cases and four-

TABLE V
SERVICE-FACING LAYER AGREEMENT. PARENTHESES GIVE UNIT COUNTS.

Allocation	Island/arch.	Coastal mainland	Difference
Equal share	0.380 (67)	-0.081 (205)	0.461
Projection score	0.632 (67)	0.001 (202)	0.631
Top-1	0.820 (67)	0.105 (202)	0.715

class counts establish cross-layer heterogeneity and show why the audit reports both continuous measures and categorical outcomes.

B. Layer Agreement Differs by Geography

Within-unit differences raise a second question: does network concentration at least preserve the ordering of corridor concentration across units? Under projection-score weighting, Layer Agreement is 0.632 across 67 island and archipelagic service-facing units and 0.001 across 202 coastal-mainland units (Table V). These units represent 10 and 28 countries, respectively, and the observed coefficient difference is 0.631. The island and archipelagic group exhibits a stronger positive concentration-rank association, while the coastal-mainland group exhibits almost no monotonic rank association. Land-locked units form a separate geographic category.

C. Geographic Ordering across Candidate Allocations

The geographic contrast retains the same direction under all three candidate-allocation rules. As allocation moves from equal shares to projection-score weighting and then Top-1, the island coefficient increases from 0.380 to 0.632 and 0.820, while the coastal-mainland coefficient changes from -0.081 to 0.001 and 0.105. The between-group difference consequently increases from 0.461 to 0.631 and 0.715. The equal-share result contains three additional coastal-mainland DNS units relative to the projection-score and Top-1 comparisons. Across

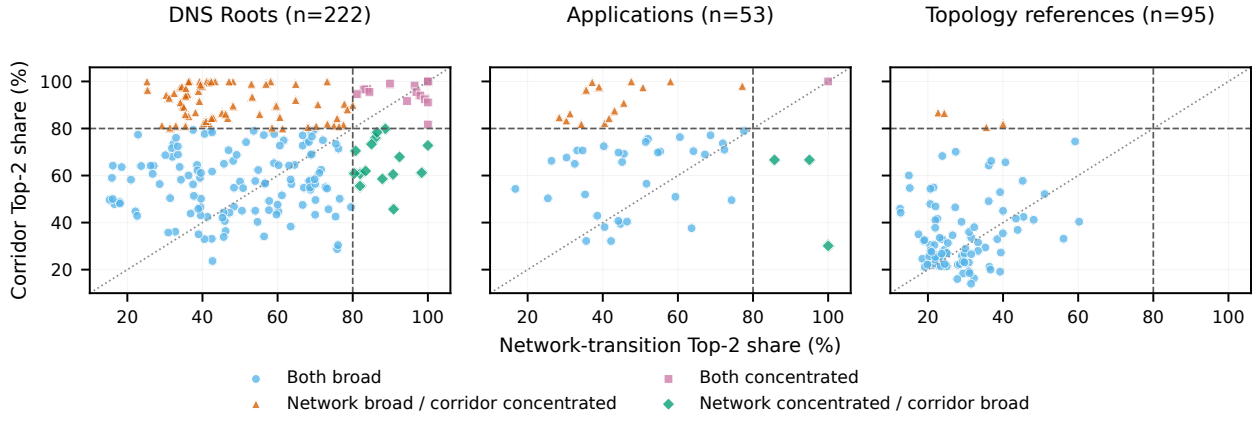


Fig. 5. Equal-share network-transition and corridor Top-2 shares for 370 auditable units. The 80% lines define the descriptive four-class view; the diagonal separates increases from decreases in concentration.

these allocation-specific cohorts, the island coefficient exceeds the coastal-mainland coefficient under equal, graded, and Top-1 support. Equal sharing preserves this ordering while assigning corridor mass independently of score magnitudes.

The reproducible equal-share result supports a country-level uncertainty check. Its 67 island and archipelagic units represent 10 countries, and its 205 coastal-mainland units represent 30 countries. Country-cluster bootstrap 95% intervals are $[-0.203, 0.760]$ for the island coefficient, $[-0.284, 0.238]$ for the coastal-mainland coefficient, and $[-0.213, 0.902]$ for their 0.461 difference. A two-sided country-label permutation gives $p = 0.161$. The wide interval for the difference includes zero. This check quantifies uncertainty across 10 island and 30 coastal country clusters; the three allocation rules separately compare the observed ordering under alternative candidate-mass assignments.

D. Inter-Region Candidate Exposure

Inter-region candidates occur infrequently for most service-facing populations. The DNS family contains 945 country-measurement units with at least 30 valid traceroutes across 77 countries. Pooling eligible Root observations within each country by valid-trace count yields median DNS exposure of 5.50% (IQR 1.10–23.04%; mean 17.80%). The difference between the median and mean records a pronounced upper tail across countries.

The three application measurements occupy the same low-exposure range. Median country-measurement exposure is 1.90% for Wikipedia, 1.96% for Reddit, and 1.43% for Netflix Assets. The dynamic multi-target topology references reach 23.24% for MSM 5051 and 23.81% for MSM 5151. These figures describe how often paths in each observed population contain a feasible inter-region corridor; they do not describe concentration after entry.

Country-matched comparisons describe exposure differences between service and topology-reference populations with the same source-country identity. DNS exposure is lower than MSM 5051 in 70 of 77 shared countries, with a median

paired difference of -14.52 percentage points. It is lower than MSM 5151 in 54 of 59 shared countries, with a median difference of -14.82 points. The paired direction holds for 90.9% and 91.5% of shared countries, respectively (Fig. 2).

E. Absolute Exposure and Breadth by Geography

The geographic groups also differ in how often DNS paths enter the candidate space. Giving each country equal weight, island and archipelagic countries have median country-pooled DNS exposure of 49.61% across 9 countries, compared with 7.88% for 54 other coastal countries and 1.70% for 14 landlocked countries. These values record how frequently each geographic group enters the feasible-corridor candidate space.

Conditional equal-share corridor breadth follows a different ordering. Island and archipelagic countries have median corridor Top-2 share of 53.3% and 5.58 effective corridors. Other coastal countries reach 75.4% and 3.39. The two represented landlocked countries reach 80.2% and 2.51. Across 38 countries with both pooled exposure and an auditable corridor summary, exposure has Spearman association -0.29 with corridor Top-2 share and 0.30 with effective corridor count. These quantities show that frequent entry and conditional concentration are not interchangeable.

F. Layer Agreement within Measurement Families

Family-level coefficients are substantially smaller than the island and archipelagic coefficient. Table VI reports Layer Agreement within each observed measurement family. Under projection-score weighting, the coefficient is 0.176 for 219 DNS units, 0.229 for 53 application units, and 0.012 for 95 topology-reference units. Equal-share and Top-1 allocation change all three coefficients, with the largest change occurring for DNS Roots. The equal-share audit contains 222 DNS units, whereas the family allocation comparison contains 219; the tables retain these allocation-specific counts.

Absolute equal-share corridor breadth also differs among the observed families. Table VII reports network and corridor medians. DNS Roots and applications have median corridor

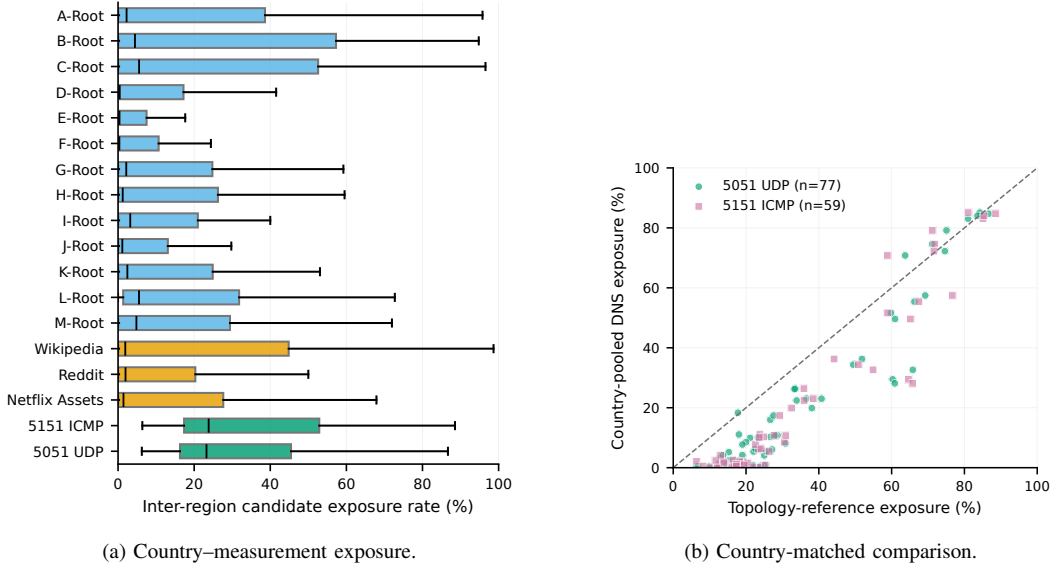


Fig. 2. Inter-region candidate exposure in the observed measurement populations.

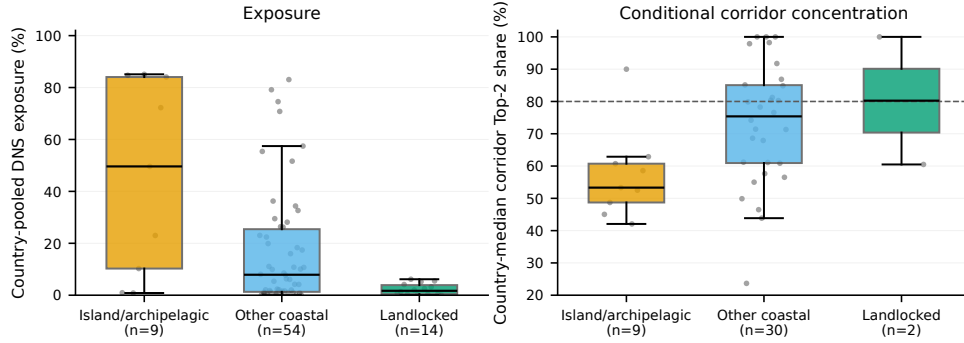


Fig. 3. DNS exposure and equal-share corridor concentration by geography. Country counts refer to the DNS country-level summary and are distinct from the service-facing unit counts in Table V.

TABLE VI
LAYER AGREEMENT BY MEASUREMENT FAMILY.

Family	n	Equal	Score	Top-1
DNS Roots	219	0.096	0.176	0.330
Applications	53	0.030	0.229	0.372
Topology references	95	0.139	0.012	0.122

Top-2 shares of 72.0% and 70.3%, compared with 30.6% for topology references. Their median effective corridor counts are 3.84 and 3.51, compared with 18.01. The two topology-reference measurements deliberately sample broader target populations, providing a breadth reference for the service-facing distributions.

The high-concentration tail follows the same descriptive family ordering. The leading two corridors receive at least 80% of mass in 83 of 222 DNS units (37.4%), 16 of 53 application units (30.2%), and 4 of 95 topology-reference units (4.2%). Wikipedia, Reddit, and Netflix Assets have median corridor

Top-2 shares of 68.3%, 71.0%, and 71.6%, with 4.25, 3.25, and 3.47 effective corridors. Their median Top-2 shifts are 11.0, 25.6, and 23.6 percentage points, respectively.

Normalized entropy supplies a support-relative view. Median entropy reduction is 0.126 for DNS Roots, 0.164 for applications, and 0.081 for topology references. The family summaries therefore agree on observed absolute breadth and concentration, while the Layer Agreement coefficients show that within-family rankings remain weak and allocation-sensitive.

VI. ROBUSTNESS, SCOPE, AND IMPLICATIONS

Section V-C reports the geographic ordering under three candidate-allocation rules and the country-clustered equal-share uncertainty check. This section examines the A-Root landing-region resolution sweep, states the measurement population, and discusses what within-unit differences and geographic rank associations imply for diversity audits.

TABLE VII
EQUAL-SHARE NETWORK-TRANSITION AND CORRIDOR DISTRIBUTIONS.

Family	Units	Net. Top-2	Corr. Top-2	Δ Top-2	Eff. network	Eff. corridor
DNS Roots	222	53.1%	72.0%	+11.0 pp	7.45	3.84
Applications	53	45.2%	70.3%	+21.0 pp	9.89	3.51
Topology references	95	27.3%	30.6%	+2.7 pp	24.34	18.01

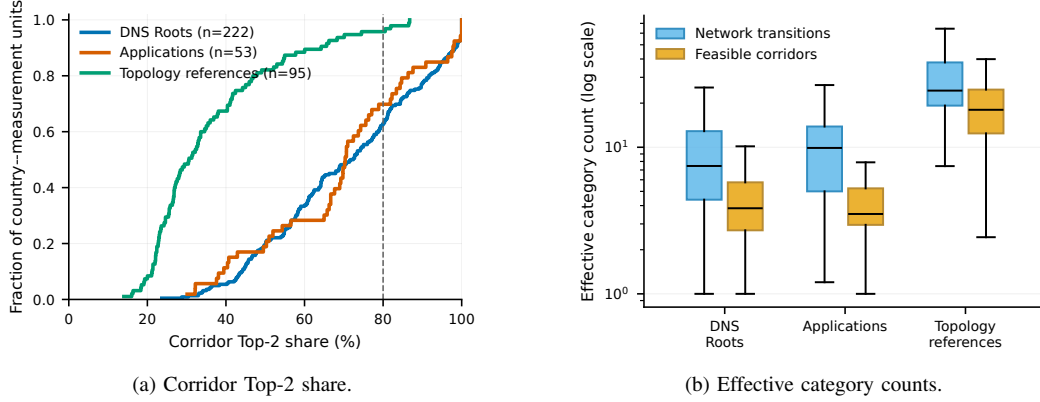


Fig. 4. Equal-share corridor concentration and breadth by measurement family. The families contain different target populations.

TABLE VIII
A-ROOT LANDING-REGION DIAMETER SENSITIVITY.

Diameter	Single-corridor	Median Top-2	Units
10 km	47.4%	53.6%	16
20 km	46.2%	62.8%	15
30 km	54.8%	64.4%	15
40 km	90.1%	81.6%	13
50 km	94.8%	84.9%	13

A. Landing-Region Resolution

The A-Root landing-region analysis varies maximum diameter over 10, 20, 30, 40, and 50 km while holding the endpoint catchment at 50 km and the RTT tolerance at 5 ms. Table VIII and Fig. 6 report the auditable-unit count, single-corridor share, and median Top-2 share. Exact landing-pair and cable candidate sets have mean Jaccard 1.0 on segments shared with the 30-km baseline. The sweep records changes in region grouping, the candidate-bearing population, and auditable-unit counts within A-Root.

The sweep shows a clear resolution transition. The 10–30-km settings retain a finer coastal partition, whereas single-corridor share rises to 90.1% and 94.8% at 40 and 50 km and median Top-2 share reaches 81.6% and 84.9%. We use 30 km because it remains on the fine side of this observed transition while consolidating nearby facilities; 40–50 km merge substantially more candidate support into single corridors. The A-Root check therefore supplies an empirical basis for the reporting resolution within the tested A-Root subset and an operational basis for the 30-km setting used throughout the audit.

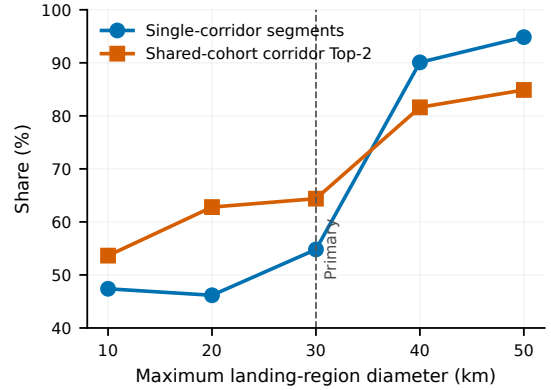


Fig. 6. A-Root landing-region diameter sensitivity under a fixed 50-km endpoint catchment and 5-ms RTT tolerance.

B. Dependence on Geolocation and Candidate Construction

Every projection begins with IPinfo hop geolocation and the fixed 50-km endpoint catchment. Geographic, lifecycle, and propagation constraints then construct the feasible candidate set. Cable lifecycle filtering aligns the inventory with July 1, 2026; a recorded inventory refresh changed 222 of 694 cable records, including 26 planned-flag changes and 47 revised service dates. The output retains exact landing-pair, cable, corridor, and mapping-resolution identities, making each projected distribution traceable to its geolocation and inventory inputs.

C. Snapshot and Population Scope

All 18 measurements cover 00:00–01:00 UTC on July 1, 2026, and the supporting datasets are aligned to the measure-

ment date. This common observation window supports cross-sectional comparisons among services and geographic groups. The reported distributions describe the paths observed during that interval.

Reported quantities are observation-weighted over participating RIPE Atlas probes and therefore represent the probes and source ASNs present in each country-measurement population. The eligibility thresholds of 10 probes and 3 probe ASNs remove sparse units from the paired analysis.

The service measurements retain their selected instances, while the dynamic multi-target references deliberately sample many destinations during the same window. The reference distributions therefore provide a broader-target context for the service-facing populations. Anycast observations jointly reflect deployed footprint, BGP selection, interconnection, and source- network context.

D. Implications for Diversity Measurement

The audit separates within-unit diversity differences from the association between rankings across units. The first comparison locates populations where broad network support accompanies concentrated corridor support or where effective breadth narrows within a concentration class. Layer Agreement then measures how closely network and corridor concentration rankings track within a geographic population. Stronger rank association can coexist with substantial within-unit differences, so both measurements are needed to interpret a network-layer diversity summary.

The DNS country-level summaries put the geographic rank contrast in context. Island and archipelagic countries have a lower median corridor Top-2 share than other coastal countries (53.3% versus 75.4%) and a higher median effective corridor count (5.58 versus 3.39). Thus, the observed DNS country-level medians show broader corridor support for the island group. Across 38 countries with both exposure and corridor summaries, exposure has Spearman associations of -0.29 with corridor Top-2 share and 0.30 with effective corridor count. These statistics describe entry frequency and conditional breadth at the country level, complementing the service-facing unit-level Layer Agreement comparison.

A possible explanation for the geographic contrast is that maritime access and network organization impose partly shared constraints on concentration rankings for island and archipelagic sources. In coastal-mainland settings, terrestrial routing, domestic backhaul, and interconnection may allow network concentration to vary more independently of feasible submarine corridors. This is a mechanism hypothesis; the present comparisons measure the geographic association. Distinguishing these explanations would require evidence linking the relevant access and routing structures to individual country-measurement outcomes.

For diversity audits, the unit-level results provide concrete inspection targets: 81 service-facing populations have broad network support but concentrated corridor support, and the continuous summaries identify additional narrowing among the 158 populations classified as broad at both layers. The

geographic comparison adds context to interpreting network-layer rankings: they track corridor concentration more closely in the observed island and archipelagic group than in the coastal-mainland group. Together, these results support checking physical-corridor concentration alongside network diversity when evaluating service paths.

Corridor concentration can be combined with cable multiplicity within a corridor, landing-station redundancy, shared terrestrial backhaul, correlated hazards, available rerouting capacity, and repair processes in a broader resilience assessment. CLASP supplies the cross-layer concentration component and preserves the exact candidate identities needed for that follow-on analysis.

E. Ethics and Reproducibility

The study uses archived public RIPE Atlas measurements and infrastructure metadata and reports aggregate country-measurement statistics. It does not target individual users or report individual probe addresses. Candidate assignments are feasible hypotheses and should not be interpreted as disclosure of an operator's actual cable use.

The released measurement IDs, configuration values, mapping-resolution states, pipeline counts, and scripts reconstruct the primary tables and figures from versioned inputs. The paper's 80% four-class rule is the primary descriptive classification; the repository's tier-based auxiliary classification uses a different rule and must remain separately named in the artifact.

VII. CONCLUSION

CLASP compares network-transition and feasible-corridor distributions over the same traceroute segments. In the aligned one-hour RIPE Atlas snapshot, 81 of 275 service-facing units have broad network support and concentrated corridor support under equal-share allocation, while effective category count contracts in 204 units (74.2%). Across units, projection-score Layer Agreement is 0.632 for 67 island and archipelagic units and 0.001 for 202 coastal-mainland units. The geographic ordering persists under equal-share and Top-1 allocation, with between-group differences of 0.461, 0.631, and 0.715 across the three rules. Network-layer diversity audits therefore need to examine both within-unit physical-corridor concentration and the geographic variation in cross-layer rank association.

VIII. USE OF AI DISCLOSURE

OpenAI ChatGPT and Codex assisted with code development, plotting scripts and language revision. Their contribution was limited to these tasks, and the authors validated the outputs against the source data and analysis results.

REFERENCES

- [1] TeleGeography, "Submarine cable map," <https://www.submarinecablemap.com/>, 2026.
- [2] R. Durairajan, S. Ghosh, X. Tang, P. Barford, and B. Eriksson, "Internet atlas: A geographic database of the internet," in *Proceedings of the 5th ACM workshop on HotPlanet*, 2013, pp. 15–20.

- [3] R. Durairajan, P. Barford, J. Sommers, and W. Willinger, "Intertubes: A study of the us long-haul fiber-optic infrastructure," in *Proceedings of the 2015 ACM conference on special interest group on data communication*, 2015, pp. 565–578.
- [4] S. Anderson, L. Salamatian, Z. S. Bischof, A. Dainotti, and P. Barford, "igdb: connecting the physical and logical layers of the internet," in *Proceedings of the 22nd ACM Internet Measurement Conference*, 2022, pp. 433–448.
- [5] A. Ramanathan and S. Abdu Jyothi, "Nautilus: A framework for cross-layer cartography of submarine cables and ip links," pp. 101–102, 2024.
- [6] C. Wang, Y. Zhang, Q. Dong, E. Carisimo, R. Durairajan, and F. E. Bustamante, "Threading the ocean: Mapping digital routes across submarine cables using calypso," in *Proceedings of the ACM SIGCOMM 2025 Conference*, 2025, pp. 1260–1262.
- [7] A. Ramanathan, R. Sankaran, and S. Abdu Jyothi, "Xaminer: An internet cross-layer resilience analysis tool," *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 8, no. 1, pp. 1–37, 2024.
- [8] S. Liu, Z. S. Bischof, I. Madan, P. K. Chan, and F. E. Bustamante, "Out of sight, not out of mind: A user-view on the criticality of the submarine cable network," in *Proceedings of the ACM Internet Measurement Conference*, 2020, pp. 194–200.
- [9] D. Cicalese, J. Augé, D. Joumlatt, T. Friedman, and D. Rossi, "Characterizing ipv4 anycast adoption and deployment," in *Proceedings of the 11th ACM Conference on Emerging Networking Experiments and Technologies*, 2015, pp. 1–13.
- [10] W. B. De Vries, R. de O. Schmidt, W. Hardaker, J. Heidemann, P.-T. de Boer, and A. Pras, "Broad and load-aware anycast mapping with verfploeter," in *Proceedings of the 2017 Internet Measurement Conference*, 2017, pp. 477–488.
- [11] H. Ye, S. Wang, and D. Li, "Impact of international submarine cable on internet routing," in *IEEE INFOCOM 2023-IEEE Conference on Computer Communications*. IEEE, 2023, pp. 1–10.
- [12] Z. S. Bischof, J. P. Rula, and F. E. Bustamante, "In and out of cuba: Characterizing cuba's connectivity," in *Proceedings of the 2015 Internet Measurement Conference*, 2015, pp. 487–493.
- [13] Z. S. Bischof, R. Fontugne, and F. E. Bustamante, "Untangling the world-wide mesh of undersea cables," in *Proceedings of the 17th ACM workshop on hot topics in networks*, 2018, pp. 78–84.
- [14] R. Fanou, B. Huffaker, R. Mok, and K. C. Claffy, "Unintended consequences: Effects of submarine cable deployment on internet routing," in *International Conference on Passive and Active Network Measurement*. Springer, 2020, pp. 211–227.
- [15] E. Carisimo, C. J. Wang, M. Weaver, F. E. Bustamante, and P. Barford, "A hop away from everywhere: A view of the intercontinental long-haul infrastructure," *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 7, no. 3, pp. 1–26, 2023.
- [16] RIPE NCC, "RIPE Atlas," <https://atlas.ripe.net/>, 2026.
- [17] IPinfo, "IP geolocation and ASN databases," <https://ipinfo.io/>, 2026.
- [18] CAIDA, "AS relationships dataset," <https://www.caida.org/catalog/datasets/as-relationships/>, 2026.